Python Data Science Practical Assignment

NumPy & Pandas Practice Questions

Contents

1 NumPy Practical Questions (20)

Q1. Create a NumPy array containing numbers from 1 to 50. Print the shape, size, and datatype of the array.

Q2. Generate an array of 20 random integers between 10 and 100. Find:

- Maximum value
- Minimum value
- Mean
- Median
- Standard deviation

Q3. Create a 5×5 matrix filled with zeros and replace the border elements with 1.

Q4. Generate a 4×4 identity matrix and explain its applications.

Q5. Create two matrices of size 3×3 and perform:

- Addition
- Subtraction
- Element-wise Multiplication
- Matrix Multiplication

Q6. Create an array from 1 to 100 and print:

- Even numbers
- Odd numbers
- Numbers divisible by both 3 and 5

Q7. Create a random array of size 25 and reshape it into:

- 5×5
- 1×25
- 25×1

Q8. Create a 6×6 random matrix and extract:

- First row
- Last column
- Center 2×2 matrix

Q9. Generate 50 random floating-point numbers between 0 and 1. Find all values greater than 0.7.

Q10. Create an array containing duplicate values. Remove duplicates using NumPy.

Q11. Reverse an array without using loops.

Q12. Create an array of numbers from 1 to 100 and replace all multiples of 5 with -1.

Q13. Create a 5×5 random matrix and calculate:

- Row-wise sum
- Column-wise sum

Q14. Generate a random matrix and normalize its values between 0 and 1.

Q15. Create two arrays and find common elements between them.

Q16. Create a 10×10 matrix and print:

- Main diagonal
- Secondary diagonal

Q17. Perform sorting in ascending and descending order.

Q18. Generate a random matrix and calculate Eigenvalues and Eigenvectors.

Q19. Create an array with missing values (NaN) and replace them with the column mean.

Q20. Mini Project: Simulate marks of 100 students in 5 subjects using NumPy. Calculate:

- Average marks
- Highest scorer
- Lowest scorer
- Subject-wise average
- Student ranking

2 Pandas Practical Questions (20)

Q1. Create a DataFrame of 10 students containing:

- Name
- Age
- Course
- Marks

Display the first five and last five records.

Q2. Read a CSV file and display:

- Shape
- Columns
- Data Types
- Summary Statistics

Q3. Find:

- Missing values
- Duplicate rows
- Remove duplicates

Q4. Filter students whose marks are greater than 80.

Q5. Sort the DataFrame:

- Ascending by Marks
- Descending by Age

Q6. Add a new column called **Result**.

- Pass if Marks ≥ 40
- Fail otherwise

Q7. Create a new column called **Grade** using conditions:

- A : 90+
- B : 80–89
- C : 70–79
- D : 60–69
- F : Below 60

Q8. Perform GroupBy on Department and calculate:

- Mean Marks
- Maximum Marks

- Count of Students

Q9. Merge two DataFrames:

- Student Information
- Student Fees

using Student ID.

Q10. Perform:

- Inner Join
- Left Join
- Right Join
- Outer Join

Q11. Create a pivot table showing Department vs Average Marks.

Q12. Replace missing values using:

- Mean
- Median
- Forward Fill
- Backward Fill

Q13. Convert:

- String to Date
- Date to Month
- Date to Year

Q14. Find the top 5 highest scoring students.

Q15. Apply Lambda function to increase marks by 5%.

Q16. Create a MultiIndex DataFrame using Department and Semester.

Q17. Export the DataFrame into:

- CSV
- Excel
- JSON

Q18. Use `value_counts()`, `unique()`, and `nunique()` on categorical columns.

Q19. Perform data visualization using Pandas:

- Line Plot
- Bar Plot
- Histogram

- Box Plot

Q20. Mini Project: Analyze a Sales Dataset containing:

- Product Name
- Category
- Quantity
- Price
- Date

Perform:

(a) Total Revenue
(b) Monthly Sales
(c) Best Selling Product
(d) Top 5 Customers
(e) Category-wise Sales
(f) Pivot Table
(g) GroupBy Analysis
(h) Missing Value Handling
(i) Data Visualization
(j) Export Final Report

Submission Guidelines

- Write clean and well-commented Python code.
- Use meaningful variable names.
- Display outputs for every question.
- Wherever applicable, explain the output.
- Submit a Jupyter Notebook (.ipynb).

Happy Coding!